MSG No. 45.240 $I_a ba2$ [Type IV, orthorhombic]

Table 1: Wyckoff site: `8a`, site symmetry: `..2`

No.	position	mapping
1	$[0,\ 0,\ z]$	`[1,2]`
2	$\left[0,\ 0,\ z+\frac{1}{2}\right]$	`[3,4]`
3	$\left[\frac{1}{2},\ \frac{1}{2},\ z+\frac{1}{2}\right]$	`[5,6]`
4	$\left[\frac{1}{2},\ \frac{1}{2},\ z\right]$	`[7,8]`
5	$\left[\frac{1}{2},\ 0,\ z\right]$	`[9,10]`
6	$\left[\frac{1}{2},\ 0,\ z+\frac{1}{2}\right]$	`[11,12]`
7	$\left[0,\ \frac{1}{2},\ z+\frac{1}{2}\right]$	`[13,14]`
8	$\left[0,\ \frac{1}{2},\ z\right]$	`[15,16]`

Table 2: Wyckoff site: `8b`, site symmetry: `..2'`

No.	position	mapping
1	$\left[\frac{1}{4},\ 0,\ z\right]$	`[1,10]`
2	$\left[\frac{3}{4},\ 0,\ z\right]$	`[2,9]`
3	$\left[\frac{3}{4},\ 0,\ z+\frac{1}{2}\right]$	`[3,12]`
4	$\left[\frac{1}{4},\ 0,\ z+\frac{1}{2}\right]$	`[4,11]`
5	$\left[\frac{3}{4},\ \frac{1}{2},\ z+\frac{1}{2}\right]$	`[5,14]`
6	$\left[\frac{1}{4},\ \frac{1}{2},\ z+\frac{1}{2}\right]$	`[6,13]`
7	$\left[\frac{1}{4},\ \frac{1}{2},\ z\right]$	`[7,16]`
8	$\left[\frac{3}{4},\ \frac{1}{2},\ z\right]$	`[8,15]`

Table 3: Wyckoff site: `8c`, site symmetry: `.m'.`

No.	position	mapping
1	$\left[x,\ \frac{1}{4},\ z\right]$	`[1,16]`
2	$\left[-x,\ \frac{3}{4},\ z\right]$	`[2,15]`
3	$\left[-x,\ \frac{1}{4},\ z+\frac{1}{2}\right]$	`[3,14]`
4	$\left[x,\ \frac{3}{4},\ z+\frac{1}{2}\right]$	`[4,13]`
5	$\left[x+\frac{1}{2},\ \frac{3}{4},\ z+\frac{1}{2}\right]$	`[5,12]`
6	$\left[\frac{1}{2}-x,\ \frac{1}{4},\ z+\frac{1}{2}\right]$	`[6,11]`
7	$\left[\frac{1}{2}-x,\ \frac{3}{4},\ z\right]$	`[7,10]`
8	$\left[x+\frac{1}{2},\ \frac{1}{4},\ z\right]$	`[8,9]`

Table 4: Wyckoff site: `16d`, site symmetry: `1`

No.	position	mapping
1	$[x,\ y,\ z]$	`[1]`
2	$[-x,\ -y,\ z]$	`[2]`
3	$\left[-x,\ y,\ z+\frac{1}{2}\right]$	`[3]`
4	$\left[x,\ -y,\ z+\frac{1}{2}\right]$	`[4]`
5	$\left[x+\frac{1}{2},\ y+\frac{1}{2},\ z+\frac{1}{2}\right]$	`[5]`

continued ...

Table 4

No.	position	mapping
6	$\left[\frac{1}{2}-x,\ \frac{1}{2}-y,\ z+\frac{1}{2}\right]$	[6]
7	$\left[\frac{1}{2}-x,\ y+\frac{1}{2},\ z\right]$	[7]
8	$\left[x+\frac{1}{2},\ \frac{1}{2}-y,\ z\right]$	[8]
9	$\left[x+\frac{1}{2},\ y,\ z\right]$	[9]
10	$\left[\frac{1}{2}-x,\ -y,\ z\right]$	[10]
11	$\left[\frac{1}{2}-x,\ y,\ z+\frac{1}{2}\right]$	[11]
12	$\left[x+\frac{1}{2},\ -y,\ z+\frac{1}{2}\right]$	[12]
13	$\left[x,\ y+\frac{1}{2},\ z+\frac{1}{2}\right]$	[13]
14	$\left[-x,\ \frac{1}{2}-y,\ z+\frac{1}{2}\right]$	[14]
15	$\left[-x,\ y+\frac{1}{2},\ z\right]$	[15]
16	$\left[x,\ \frac{1}{2}-y,\ z\right]$	[16]